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Markov Regime-Switching Model for Options Pricing

Methodology Overview — Code Sample Context Document

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Code repository: github.com/gregorybarco/OAG_OptionsPricing_Snippet

Interactive visualizations: BarcoGregory.com/oag_application_sample

ABSTRACT

This document provides context for the accompanying Python code sample submitted in support of OAG application RAD_NYC_DAT_6444. The code demonstrates data ETL via a live brokerage API, quantitative analysis, and production-grade data visualization. It is drawn from an in-progress research manuscript co-authored with PhD faculty at Brooklyn College, CUNY, developing a system for pricing and calibrating equity options under a two-state Markov regime-switching model.

WHAT THE CODE SAMPLE CONTAINS

The repository contains three Python modules, each directly relevant to the data analyst role: **(1) ETL Pipeline** — programmatic extraction, authentication, and structured parsing of live options chain data from the Charles Schwab brokerage API; **(2) Contract Selector** — filtering and aggregation logic that identifies relevant contracts by expiry, strike, and liquidity; **(3) Fitness Landscape Scanner** — analytical visualization tool that maps how well any given set of model parameters fits the observed market data, producing interactive 3D surfaces with Plotly.

MOTIVATION AND ANALYTICAL APPROACH

Financial markets alternate between structurally distinct environments. A model assuming constant volatility systematically misprices risk during high-volatility regimes. By detecting when markets switch states, the system flags structural changes in market behavior — analogous to detecting anomalies in a data stream — before those changes are visible in price levels alone. This approach is directly applicable to identifying irregular patterns in market or economic data of the kind relevant to OAG investigations.

DATA PIPELINE

Live options chains are retrieved programmatically via the Charles Schwab brokerage API across 11 equity tickers including SPX, AAPL, MSFT, and QQQ. Risk-free rates and implied dividend yields are sourced from Bloomberg Terminal. The ETL layer handles OAuth2 token lifecycle management, automatic re-authentication, JSON parsing, and structured output — all implemented in Python and available in full in the accompanying code sample.

KEY FINDINGS

Single-name equities calibrate cleanly with well-defined parameter basins. Broad index assets such as SPX produce ridge-shaped fitness surfaces, indicating the two-state structure is insufficient to capture institutional crash-hedging demand. Identifying the limits of a model is as analytically important as identifying what it explains well. Interactive visualizations of these findings are available at the portfolio link above.

COMPUTATIONAL INFRASTRUCTURE

Python 3.11 orchestrates the full pipeline. The numerical core is implemented in Fortran compiled with the NVIDIA HPC SDK, linked against Intel MKL, and parallelized via OpenMP and CUDA on an RTX 5060 Ti GPU — achieving approximately 100× speedup over CPU computation. Visualizations are produced with Plotly and Matplotlib, exported as self-contained interactive HTML.